



MAR ATHANASIOUS COLLEGE OF ENGINEERING, KOTHAMANGALAM

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

COURSE CODE : ECD334

MINI PROJECT PRESENTATION

PHASE I & II

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CONTENTS

PHASE I



- Working Principle
- Design
- Components Required
- Circuit Diagram
- Schematic Diagram
- PCB Layout
- Fabrication steps
- Output & Results
- Challenges
- Applications & Future scope

PHASE II



- Introduction
- Objectives
- Components Required
- Block Diagram
- Working Principle
- Features
- Design & Implementation
- Results
- Challenges
- Applications & Future scope

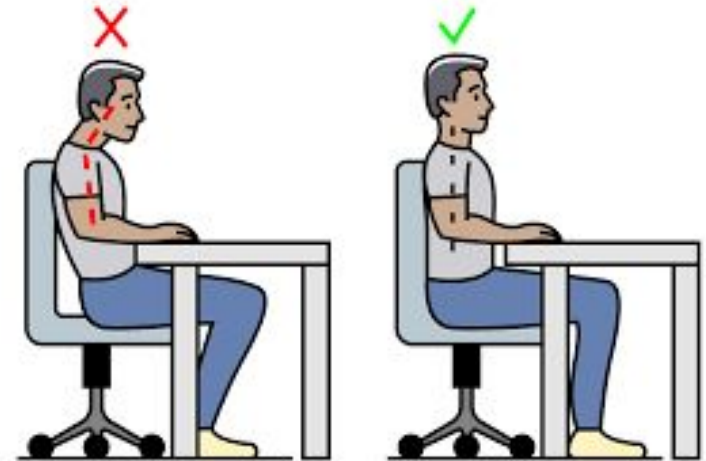
WHY POSTURE CORRECTION MATTERS ?

- Extended sitting is common among **students** and **IT professionals**
- Improper posture causes **neck pain, back pain, and muscle fatigue**
- Sustained poor posture may lead to **cervical spondylosis**, back strain, and stiffness



WHAT IS CONSIDERED GOOD POSTURE ?

- Good posture maintains a **near-neutral spine**
- A small natural curve in the back is maintained
- Posture is **personalized**, not based on a single fixed angle



OBJECTIVES



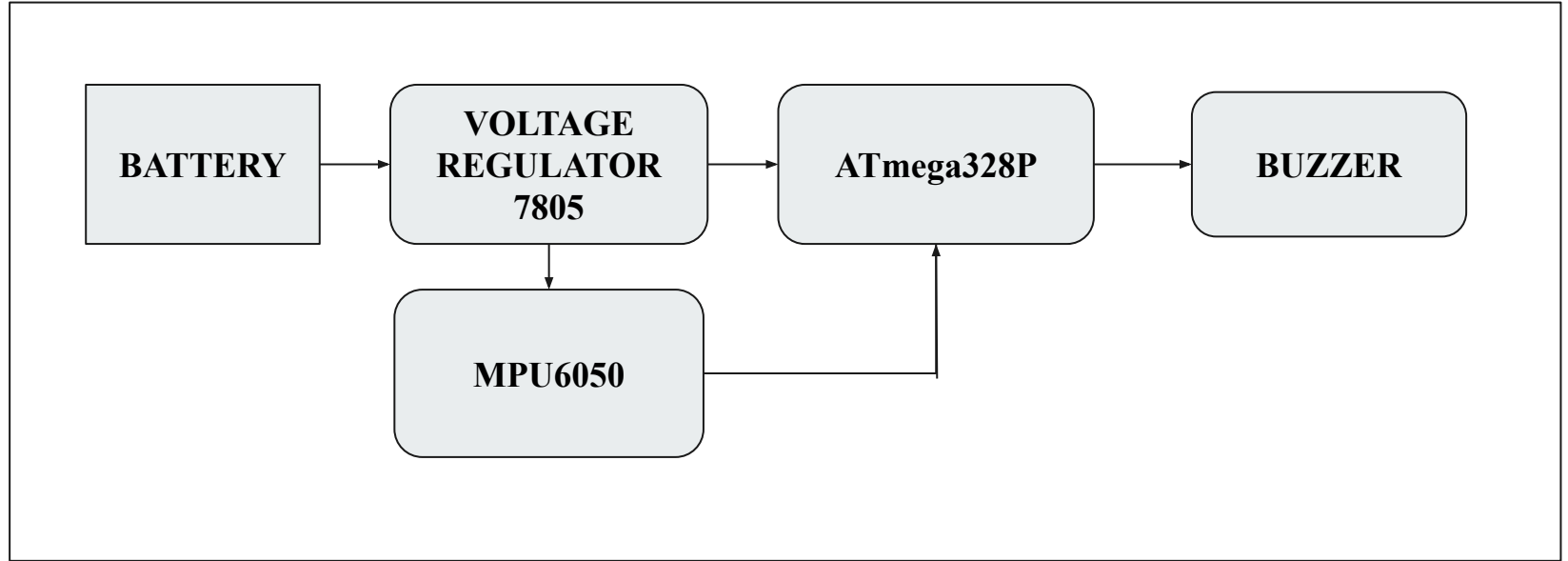
- To design a wearable posture correction system
- To implement posture detection using **MPU6050 sensor** data
- To process orientation data using the **ATmega328P** microcontroller
- To perform **baseline calibration** for **user-specific** posture adaptation
- To generate corrective feedback using a buzzer alert system
- To develop a **custom-designed PCB** for a compact embedded solution
- To validate the system through **fabricated hardware testing**

HARDWARE COMPONENTS



- ATmega328P microcontroller
- MPU6050 sensor
- Buzzer
- BC547 transistor
- 7805 voltage regulator
- 16 MHz crystal oscillator
- capacitors, resistors, reset switch
- FTDI and ICSP programming headers
- 9V power input

BLOCK DIAGRAM



WORKING PRINCIPLE



1. Device powers on → a startup beep confirms readiness
2. Good posture is recorded as **baseline pitch**
3. Sensor continuously measures current tilt
4. Controller computes **corrected pitch** = current pitch – baseline
5. If slouch exceeds threshold for a **3 seconds**, buzzer sounds
6. If posture becomes worse, **beep rate increases**
7. When posture is corrected, alert stops

WORKING PRINCIPLE

MPU6050 communicates over **I²C** and provides:

- **Accelerometer output as** linear acceleration along X, Y, Z axes (in m/s²)
- Converted into **angle** (degrees)

$$\text{Pitch} = \tan^{-1} \left(\frac{-a_y}{a_z} \right) \times (180 / \pi)$$

Where a_y and a_z are acceleration components

FEATURES



1. Baseline Calibration

- On startup, the user is prompted: "Hold good posture for calibration"
- After a **2-second** settle delay, one accelerometer reading is captured
- This pitch value becomes **baseline pitch** — stored for the entire session
- All subsequent readings are subtracted from this baseline

$\text{corrected pitch} = \text{current pitch} - \text{baseline pitch}$

FEATURES



2. Decision Logic and Adaptive Alert System

CONDITION	ANGLE	BEEP INTERVAL
Good Posture	$<14^{\circ}$	No Beep
Bad Posture	$18-30^{\circ}$	Every 2 seconds
Very Bad Posture	$>30^{\circ}$	Every 700 ms

Beep frequency: 2000 Hz, duration: 120 ms

FEATURES



3. False Alert Prevention

- Short-term bending (like picking objects) should not trigger alert
- Solution implemented:

3-second delay before alert

Neutral zone ($<5^\circ$) resets system

Filtering removes sudden spikes

- Only continuous bad posture triggers buzzer

SIMULATION USING WOKWI

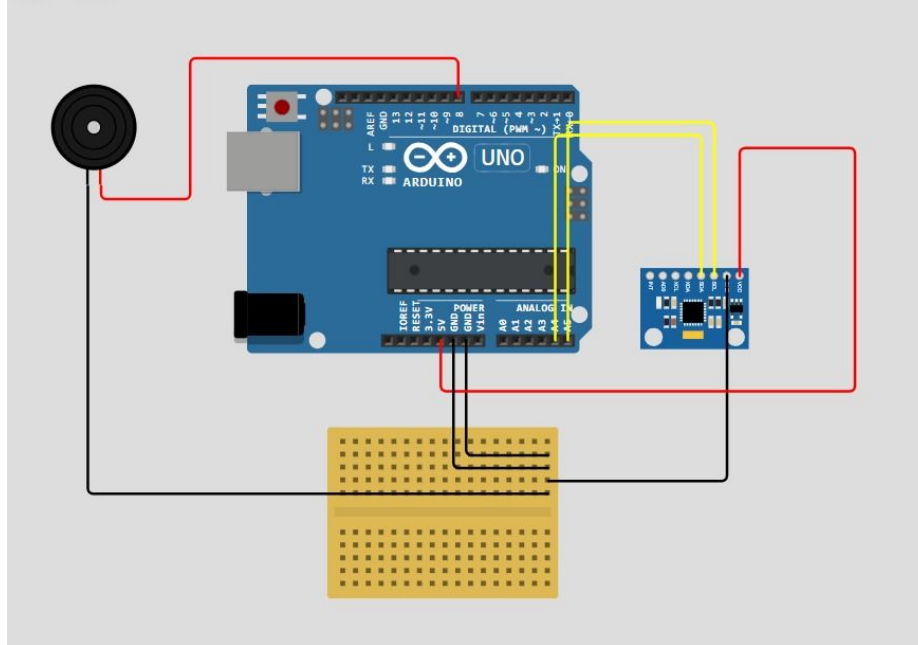


FIG 1: POSTURE CORRECTOR IMPLEMENTED USING ARDUINO UNO

SCHEMATIC DESIGN USING ALTIUM

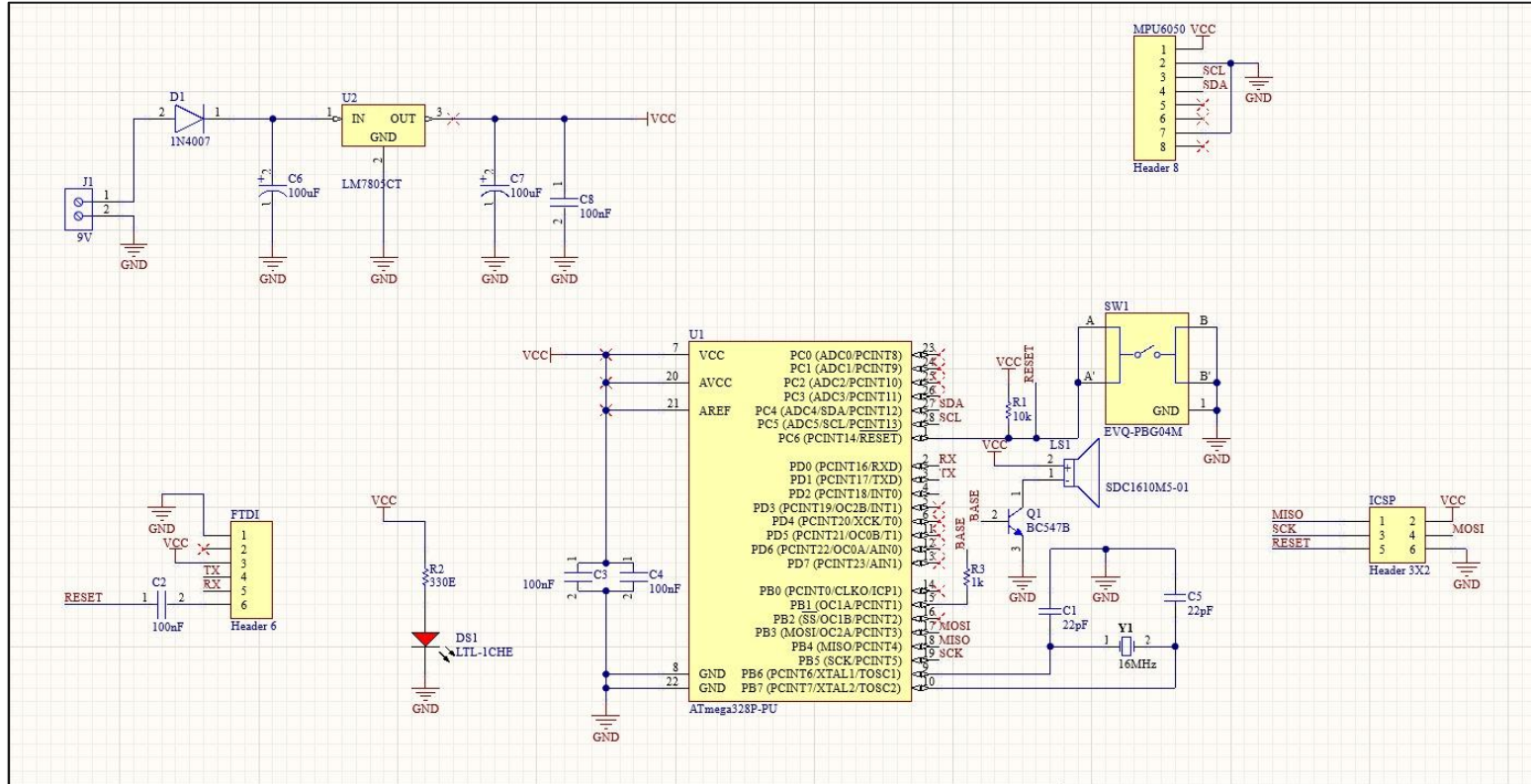


FIG 2 : SCHEMATIC DESIGN OF POSTURE CORRECTOR USING ATMEGA328P

CIRCUIT OVERVIEW



1. Power Section

Regulated 5V Power Supply

- **Input:** 9V external supply
- **1N4007 diode** — reverse polarity protection, prevents damage if battery is connected backwards
- **LM7805** — linear voltage regulator, steps down 9V → stable 5V
- **100 μ F capacitor** — bulk filtering, absorbs low-frequency voltage fluctuations
- **100nF capacitor** — high-frequency noise decoupling, keeps supply clean at the IC pins

CIRCUIT OVERVIEW



2. Microcontroller Core

ATmega328P

- Heart of the system — runs all posture logic, timing, and I²C communication
- 16 MHz crystal (Y1) — provides precise clock reference
- 22pF capacitors (C1, C5) — load capacitors for crystal stability

CIRCUIT OVERVIEW



2. MPU6050 Interface

- Communicates with ATmega328P over I²C protocol (2-wire)
- SDA → PC4 (Arduino A4) — Serial Data line
- SCL → PC5 (Arduino A5) — Serial Clock line
- Powered directly from the board's regulated VCC and GND

CIRCUIT OVERVIEW



4. Programming Interfaces

- Two Ways to Program the Board
- Having both makes the board flexible for all stages of development

INTERFACE	PINS USED	PURPOSE
FTDI (UART)	TX,RX	Upload sketches via USB-to-Serial adapter
ICSP (SPI)	MOSI, MISO, SCK, RESET	Burn bootloader

CIRCUIT OVERVIEW



5. Buzzer Drive Circuit — BC547 Transistor

- ATmega328P GPIO can only source $\sim 40\text{mA}$ — not enough to drive a buzzer directly
- BC547 NPN transistor acts as a switch, allowing the buzzer to draw current from VCC
- Base resistor (R3) — limits base current, protects the microcontroller's GPIO pin
- Buzzer (LS1) — driven safely through the collector with full supply voltage

PCB Implementation

- Custom standalone ATmega328P design
- I2C connection to MPU6050
- Transistor-driven buzzer
- Onboard reset and programming support
- Compact PCB for final implementation

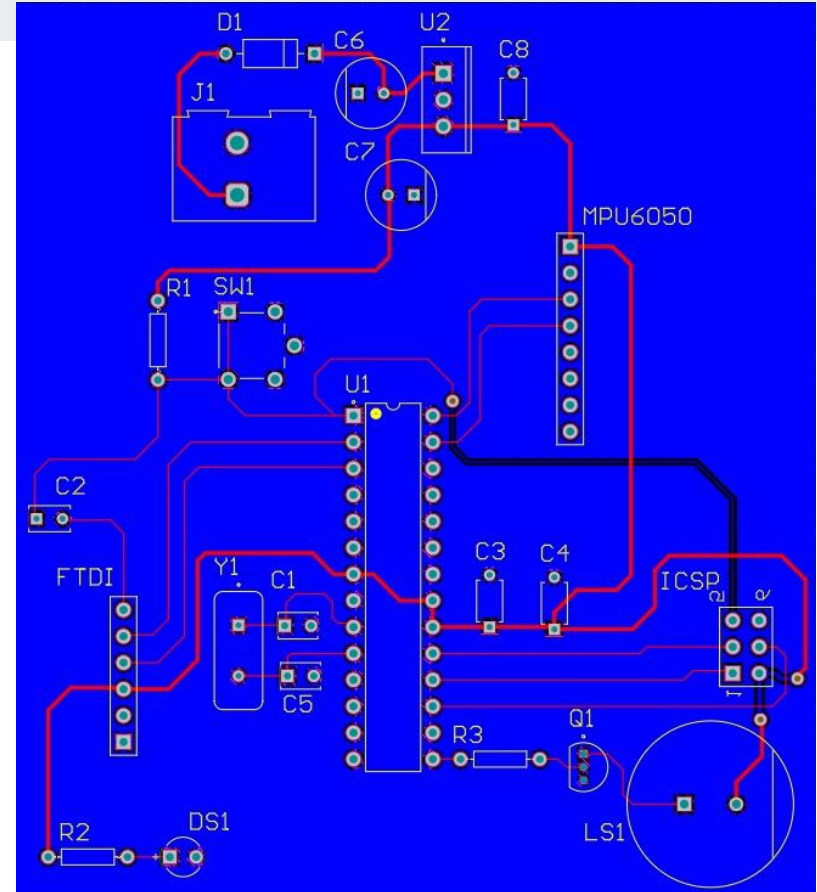


FIG 3 : PCB LAYOUT USING ALTIUM

FABRICATED PCB



HARDWARE VALIDATION



- Verified I2C communication with MPU6050
- Observed real-time angle via Serial Monitor
- Tested:

Straight posture → no beep

Slouch → delayed alert

Severe slouch → fast alert

Existing Wearables vs Our Approach



TYPICAL WEARABLE POSTURE CORRECTOR	SMART POSTURE CORRECTOR
• Usually Strap-Based Or Brace-Based • Physically Force The Shoulders/Back Into Position • Can Feel Uncomfortable • May Create Dependence Instead Of Awareness • Not Suitable For Long-Term Use	• Does Not Force Posture Mechanically • Monitors Posture Electronically • Gives Feedback Only When Needed • Supports Habit Formation Through Repeated Feedback • More Like Training The User Than Restraining The User

CHALLENGES FACED



- Pad sizes on certain footprints were undersized, making clean solder joints harder to achieve for some components.
- Raw accelerometer output contained high-frequency noise, causing the calculated pitch angle to fluctuate continuously → Implemented a low-pass filter in software to smooth readings

LIMITATIONS



- **Sensitive to movement:** Accurate mainly when seated; walking/dynamic motion reduces pitch accuracy.
- **No data logging:** Operates in real-time only; no history or progress tracking available.

APPLICATIONS



- Students and office workers
- Rehabilitation Therapy
- Fitness and posture training
- Long duration computer users

FUTURE SCOPE



- Vibration motor instead of buzzer
- Rechargeable battery
- Smaller wearable enclosure
- Mobile app integration
- Posture logging and analytics

THANK YOU